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DETAIL SPECIFICATION

MOUNTS, RESILIENT, MARE ISLAND TYPES 11M15, 11M25 AND 10M50

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Mare Island Rubber Laboratory type resilient mounts used to support equipment on surface ships and submarines. This specification includes requirements and verification tests for the rubber compound and fully assembled mounts.

1.2 Classification.

1.2.1 Types (see 6.2).

11M15 – Nominal resonant frequency of 11 Hertz at 15 pounds with a load range from 9 to 15 pounds.

11M25 – Nominal resonant frequency of 11 Hertz at 25 pounds with a load range from 15 to 25 pounds.

10M50 – Nominal resonant frequency of 10 Hertz at 50 pounds with a load range from 25 to 50 pounds.

1.2.1.1 Non-magnetic designation. The suffix "NM", following the mount type, designates non-magnetic construction.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items

(Copies of this document are available online at http://quicksearch.dla.mil/ or https://assist.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385824 - Mounts, Resilient Mare Island Type

(Copies of this document are available from the applicable repositories listed in NAVSEA S0005-AE-PRO-010/EDM. Copies of NAVSEA S0005-AE-PRO-010/EDM are available from the Naval Logistics Library, 5450 Carlisle Pike, Mechanicsburg, PA 17055 or online at https://nll.ahf.nmci.navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D395 - Standard Test Methods for Rubber Property – Compression Set

ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers – Tension

ASTM D429 - Standard Test Methods for Rubber Property – Adhesion to Rigid Substrates

ASTM D471 - Standard Test Method for Rubber Property – Effect of Liquids

ASTM D573 - Standard Test Method for Rubber – Deterioration in an Air Oven

ASTM D792 - Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement

ASTM D1229 - Standard Test Method for Rubber Property – Compression Set at Low Temperatures

ASTM D1566 - Standard Terminology Relating to Rubber

ASTM D2240 - Standard Test Method for Rubber Property – Durometer Hardness

ASTM D5992 - Standard Guide for Dynamic Testing of Vulcanized Rubber and Rubber-Like Materials Using Vibratory Methods

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Mounts furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Mount design. Mounts shall meet requirements of this specification and be constructed in accordance with 803-1385824. In addition to requirements contained in this specification, typical static load-deflection, radial dynamic stiffness, and damping performance are provided for guidance in 6.5.

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3.4 Materials.

3.4.1 Metal. Mount metal components shall be manufactured from steel or manganese bronze as specified (see 6.2). Metal shall be in accordance with the appropriate specification listed in 803-1385824; substitutions are permitted when approved by NAVSEA. Components shall be formed into shape and finished in accordance with the dimensions and allowable tolerances specified in 803-1385824.

3.4.2 Rubber.

3.4.2.1 Resilient element. The resilient element of finished mounts shall be fabricated of oil resistant compounds utilizing polymerized chloroprene as the basic material and molded into the form as shown in 803-1385824.

3.4.2.2 Physical requirements. Physical properties of the cured rubber compound used in the resilient element shall meet all requirements specified in table I.

TABLE I. Physical requirements.

Property	Mount type		Verification method
	11M15 and 11M25	10M50	
1. Tensile strength (minimum) – before aging	2200 psi	2800 psi	4.4.1.1
2. Ultimate elongation (minimum) – before aging	650%	575%	4.4.1.1
3. Tensile strength (minimum) – after oven aging at 194 °F	2100 psi	2700 psi	4.4.1.1 and 4.4.1.7
4. Ultimate elongation (minimum) – after oven aging at 194 °F	600%	525%	4.4.1.1 and 4.4.1.7
5. Compression set (maximum) – after oven aging at 194 °F	40%	34%	4.4.1.2 and 4.4.1.7
6. Cold compression set (maximum)	50%	60%	4.4.1.3
7. Hardness	38±5	48±5	4.4.1.4
8. Volume change in oil (maximum)	15%		4.4.1.5
9. Adhesion to metal (minimum)	450 psi	550 psi	4.4.1.6
10. Specific gravity	Within 0.02 of the value measured during qualification		4.4.1.8

3.5 Performance and construction requirements for finished mounts.

3.5.1 Construction requirements. Mounts shall be constructed in accordance with 803-1385824. In addition, there shall be no backrinding, blisters, tears, cracks, or other defects on the outer surface of the resilient element. MIL-STD-407, in conjunction with ASTM D1566, may be used to assist in identifying defects in the resilient element (see 4.3.3.1).

3.5.2 Dynamic stiffness. The dynamic stiffness of mounts, when tested in the axial direction in accordance with 4.4.2.1, shall be within the specified limits of table II.

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TABLE II. Axial dynamic stiffness requirements.

Mount type	Requirement limits	
	Axial load (lb)	Dynamic stiffness (lb/in)
11M15	9	190 to 230
	12	181 to 224
	15	166 to 214
11M25	15	283 to 398
	18	278 to 403
	22	269 to 404
	25	260 to 403
10M50	25	459 to 590
	33	438 to 562
	42	429 to 557
	50	414 to 553

3.5.3 Static strength. When tested in accordance with 4.4.2.2, mounts shall not show any separation, break in or between component parts, debonding, or permanent deformation of metal parts.

3.5.4 Fatigue-drift. When tested in accordance with 4.4.2.3, mounts shall not show any indication of damage to the resilient element, rubber-to-metal bond, and metal parts. The drift occurring to the type 11M15, 11M25, and 10M50 mounts during these tests shall not exceed 0.017, 0.025, and 0.040 inch, respectively.

3.5.5 Delamination and porosity. When tested in accordance with 4.4.2.4, the rubber shall not show evidence of porosity, separation into distinct layers, or laminations.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. The inspection and tests specified herein are required to qualify mounts produced by a manufacturer for inclusion in the Qualified Products List (QPL) for this specification. Refer to 3.1 and 6.3. Qualification inspection shall be performed on finished mounts and the rubber compound.

4.2.1 Qualification inspection of the rubber compound. Specimens described in table III shall be produced from the rubber compound which is mixed for the manufacture of all mounts subjected to qualification inspection. The manufacturer shall certify that the specimens are of the same material and equivalent cure as the mounts tested for qualification. These specimens shall be subjected to the tests specified in table IV to determine compliance with 3.4.2.2. If any specimens tested fail to meet any requirement, all mounts manufactured for qualification from rubber batches represented by that specimen shall be rejected.

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TABLE III. Rubber specimens for qualification.

Quantity	Specimen description	Property to be measured
6	ASTM D412, Die C from sheets 0.08±0.01 by 6 by 6 inches	Tensile strength and ultimate elongation (initial and aged), specific gravity
3	Rectangular: 0.08±0.01 by 1 by 2 inches	Resistance to oil (volume change in oil)
3	ASTM D429, Method A Cylinder: 0.125±0.005-inch thickness by 1.597±0.005-inch diameter	Adhesion to metal
5	Cylinder: 0.49±0.02-inch height by 1.14±0.02-inch diameter	Hardness and compression set

TABLE IV. Qualification tests on rubber specimens.

Property	Verification method	Requirement
1. Initial tensile strength	4.4.1.1	3.4.2.2 (table I, no. 1)
2. Initial ultimate elongation	4.4.1.1	3.4.2.2 (table I, no. 2)
3. Tensile strength after oven aging	4.4.1.1 and 4.4.1.7	3.4.2.2 (table I, no. 3)
4. Ultimate elongation after oven aging	4.4.1.1 and 4.4.1.7	3.4.2.2 (table I, no. 4)
5. Compression set after oven aging	4.4.1.2 and 4.4.1.7	3.4.2.2 (table I, no. 5)
6. Cold compression set	4.4.1.3	3.4.2.2 (table I, no. 6)
7. Hardness	4.4.1.4	3.4.2.2 (table I, no. 7)
8. Volume change in oil	4.4.1.5	3.4.2.2 (table I, no. 8)
9. Adhesion to metal	4.4.1.6	3.4.2.2 (table I, no. 9)
10. Specific gravity	4.4.1.8	3.4.2.2 (table I, no. 10)

4.2.2 Qualification inspection of finished mounts. If rubber specimens meet all requirements associated with qualification inspection (see 4.2.1), then finished mounts shall be subjected to qualification inspection to determine compliance with 3.3, 3.5, and 803 1385824. Four mounts for each type manufactured shall undergo qualification testing. Two mounts shall be tested in accordance with table V. The remaining two mounts shall be tested in accordance with table VI. Mounts shall be tested in the sequence shown in each table.

TABLE V. Qualification tests on finished mounts, group one.

Property	Verification method	Requirement
1. Physical appearance, dimensions and materials	803-1385824 and table X	3.5.1
2. Dynamic stiffness	4.4.2.1	3.5.2
3. Static strength	4.4.2.2	3.5.3

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TABLE VI. Qualification tests on finished mounts, group two.

Property	Verification method	Requirement
1. Physical appearance, dimensions and materials	803-1385824 and table X	3.5.1
2. Dynamic stiffness	4.4.2.1	3.5.2
3. Fatigue-drift	4.4.2.3	3.5.4
4. Delamination and porosity	4.4.2.4	3.5.5

4.3 Conformance inspection. Conformance inspection shall be performed on each production lot.

4.3.1 Lot. For the purpose of sampling for conformance inspection, a lot shall contain all mounts of one type produced at the same time in the same facility under the same conditions with the same batch of rubber compound. A lot number shall be assigned to the mounts and shall not be repeated in any one quarter. The lot number with elastomer cure date (quarter, year) shall be traceable to the rubber batch number, manufacturing/process control records, and conformance documentation. The lot number along with cure date shall be permanently marked on each mount (refer to 803-1385824) and be included on all shipping documents, packages, and shipping containers.

4.3.2 Conformance inspection of the rubber compound. The specimens identified in table VII shall be prepared for each mount lot produced. The specimens shall be made from the same batch of rubber compound mixed for the manufacture of mounts in the lot. The specimens shall be certified to be of the same material and equivalent cure as the corresponding lot of finished mounts. These specimens shall be subjected to verification tests specified in table VIII to determine compliance with 3.4.2.2. If any specimens tested fail to meet any requirement, all mounts in the lot represented by that specimen shall be rejected.

TABLE VII. Rubber specimens for conformance inspection.

Quantity	Specimen description	Property to be measured
3	ASTM D412, Die C from sheets 0.08±0.01 by 6 by 6 inches	Tensile strength and ultimate elongation, specific gravity
1	Cylinder: 0.50±0.02 inch height by 1.14±0.02 inch diameter	Hardness

TABLE VIII. Conformance tests on rubber specimens.

Verification test	Test method	Requirement
1. Initial tensile strength	4.4.1.1	3.4.2.2 (table I, no. 1)
2. Initial ultimate elongation	4.4.1.1	3.4.2.2 (table I, no. 2)
3. Specific gravity	4.4.1.8	3.4.2.2 (table I, no. 10)
4. Hardness	4.4.1.4	3.4.2.2 (table I, no. 7)

4.3.3 Conformance inspection of finished mounts. Mounts shall be sampled and subjected to verification tests and examination in accordance with 4.3.3.1 and 4.3.3.2.

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4.3.3.1 Sampling for visual inspection. For each lot, a quantity of mounts sampled in accordance with table IX shall be subjected to a visual examination to determine compliance with 3.5.1. Classifications of defects are shown in table X. If one or more major defect is found in any sample, the manufacturer may screen the entire lot for compliance prior to proceeding with qualification testing; otherwise, the entire lot shall be rejected. Minor defects discovered on each mount shall be corrected when possible. MIL-STD-407, in conjunction with ASTM D1566, may be used to assist in identifying defects in the resilient element. Corrective action shall be taken to prevent the systematic occurrence of any defects.

TABLE IX. Sampling for visual inspection.

Number of mounts in the lot	Number of mounts in the sample
40 or under	10
41 to 110	15
111 to 300	25
301 to 500	35
501 to 800	50
801 to 1300	75
1301 to 3200	110

TABLE X. Classification of defects.

Category	Defects
Major:	
101	Evidence of unauthorized materials, including lack of documentation demonstrating components, were produced in accordance with material specifications listed on 803-1385824 and acquisition requirements.
102	Mount dimensions, materials, and construction not in accordance with 803-1385824 and acquisition requirements.
103	Evidence of delamination or air pockets in resilient element.
104	Evidence of backrinding, blisters, cracks, debonding, breaks, deep gouges, tears, voids, blisters, or other imperfections on the surface of resilient element that can effect mount performance and service life.
105	Mounts with cure dates more than 2 years old shall not be offered.
106	Any other defect that could affect mount performance or service life.
Minor:	
201	Identification marking not distinct.
202	Burrs, rough edges, and sharp corners not removed.
203	Minor surface imperfections on the rubber that will not affect mount performance or service life.
204	Any other negligible defect which would not affect the performance or serviceability of the mount.

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4.3.3.2 Sampling for conformance testing. For each lot, a quantity of mounts sampled in accordance with table XI shall be subjected to verification tests in table XII to determine compliance with 3.5.2 and 3.5.3. Tests shall be conducted in the sequence shown in table XII. For lots consisting of more than 40 mounts, a doubling sampling plan shall be used. If the number of failures found on the first sample does not exceed the maximum allowable, sampling shall be stopped and the lot shall be considered acceptable. If the number of failures found on the first sample exceeds the allowable failures in the combined sample, the sampling shall be stopped and the lot rejected. If the number of failures on the first sample exceeds the allowable number of first sample failures but does not exceed the allowable failures in the combined sample, a second sample shall be selected. If the combined number of failures found in the first and second sample do not exceed the allowable number of combined failures, the lot shall be considered acceptable.

TABLE XI. Sampling conformance tests.

Number of mounts in a lot	Sample quantities			Maximum allowable number of failures			
	First sample quantity	Second sample quantity	First and second sample quantities combined	In first sample quantity		In first and second sample quantity combined	
				For any one test	For all tests combined	For any one test	For all tests combined
2 to 8	3	Use single sampling plan for these lots		0	0	-	-
9 to 15	4			0	0	-	-
16 to 25	5			0	0	-	-
26 to 40	6			0	0	-	-
41 to 65	4	4	8	0	0	1	1
66 to 110	5	5	10	0	0	1	1
111 to 180	6	6	12	0	0	1	1
181 to 300	8	8	16	1	1	1	2
301 to 500	10	10	20	1	1	2	2
501 to 800	12	12	24	1	1	2	3
801 to 1300	15	15	30	1	2	2	3
1301 to 3200	20	20	40	2	2	3	4

TABLE XII. Conformance verification tests.

Verification test	Test method	Requirement
1. Dynamic stiffness and resonant frequency test	4.4.2.1	3.5.2
2. Static strength test in axial direction	Axial strength test shall be in accordance with 4.4.2.2.2, except the peak load obtained shall be “Level A” for all four loading-unloading cycles.	3.5.3

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4.4 Verification tests. Unless otherwise specified in a particular test procedure, all tests shall be conducted at an ambient temperature of 80±10 °F. Within this range, the temperature from the beginning to the end of any one test shall not vary more than ±5 °F.

4.4.1 Rubber, test methods.

4.4.1.1 Tensile strength and ultimate elongation tests. The tensile strength and ultimate elongation shall be measured in accordance with ASTM D412, Method A, using three Die C test specimens. Refer to table III or table VII for specimen description.

4.4.1.2 Compression set after oven aging. Compression set shall be determined on three rubber test specimens in accordance with ASTM D395, Method B, except oven aging conditions shall be as specified in 4.4.1.7. Rubber specimens with a hardness less than or equal to 44 durometer shall be compressed 40 percent. Rubber specimens with a hardness greater than or equal to 45 durometer shall be compressed 30 percent. Refer to table III or table VII for specimen description.

4.4.1.3 Cold compression set. Low temperature compression set shall be determined in accordance with ASTM D1229 on two rubber specimens, except compression shall be 40 percent and exposure shall be 30±2 °F for 94±½ hours. Two tests agreeing within 5 percent of each other shall be conducted and averaged to determine compliance with 3.4.2.2. Refer to table III or table VII for specimen description.

4.4.1.4 Hardness. The hardness of the ½-inch thick elastomer specimens shall be determined in accordance with ASTM D2240 with a Type A Shore durometer. A 3-second reading shall be taken to determine hardness. Refer to table III or table VII for specimen dimensions.

4.4.1.5 Volume change in oil. Volume change in oil shall be determined in accordance with ASTM D471, except the three specimens shall be immersed for a period of 70±¼ hours at 158±2 °F. The specimen shall be immersed in reference oil IRM 903 of ASTM D471. Refer to table III for specimen size.

4.4.1.6 Adhesion to metal tests. The adhesion tests shall be in accordance with Method A of ASTM D429. Three specimens of the rubber compound bonded to metal shall be tested and results averaged to determine compliance with 3.4.2.2. Specimens shall have the same surface preparation, adhesive system, metal type, and bonding method as used in the finished mount. Refer to table III for specimen description.

4.4.1.7 Oven aging test. Specimens for tensile, ultimate elongation, and compression set after oven aging shall be oven-aged in accordance with ASTM D573 at a temperature of 194±2 °F for 46±¼ hours. Final determination of aged tensile strengths and ultimate elongations shall be made not less than 16 hours, or more than 48 hours, after removal from the oven. Tensile strength and ultimate elongation tests on unaged specimens shall be made immediately prior to, and on the same machine as, oven-aged specimens.

4.4.1.8 Specific gravity. The specific gravity shall be determined in accordance with ASTM D792 on samples cut from the specimen sheets. Specimen sheets are referred to in table III and table VII. The average of three measurements shall be used to determine compliance with 3.4.2.2.

4.4.2 Finished mount, test methods.

4.4.2.1 Dynamic stiffness measurement.

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4.4.2.1.1 Test apparatus for measuring dynamic stiffness. The test apparatus shall be capable of superimposing cyclical vibration while statically loading the mount. Refer to the forced non-resonant section of ASTM D5992 for discussion and schematic of test systems capable of conducting low-frequency dynamic stiffness measurements. A commonly used type of equipment for non-resonant testing is a closed-loop servo-hydraulic machine or a vibration shaker with load frame (see figure 1). Transducers shall be arranged to measure force at the blocked output or foundation side of the mount and displacement at the equipment or input side of the mount. The stiffness of the test system (machine, load-frame, force transducer, etc.) in series with the mount under test shall be determined experimentally or via calculation. A test system which is at least 100 times stiffer than the dynamic stiffness of the mount being tested shall not require measurement adjustment to account for test system flexibility. If the test system fails to meet this relative stiffness requirement, the test system compliance shall be subtracted from the measurement to obtain the true dynamic stiffness of the mount. The elastic stiffness of the mount (K_{mount}) can be calculated using the classic equation governing the summation of springs in series:

$$(1/K_{measured}) = (1/K_{mount}) + (1/K_{system})$$

To avoid over-correction, the stiffness of the test system shall be greater than 30 times the stiffness of the mount. Refer to ASTM D5992 for guidelines and requirements of the mechanical apparatus and instrumentation used in the test system.

Dynamic calibration of the complete system shall be verified using steel springs of known spring constants, having negligible damping, prior to the start of testing. Refer to the “calibration for measurement of phase” section of ASTM D5992 for measurement discussion and selection of springs. In addition, verification springs shall be selected on the basis of their linear force-deflection properties and ability to produce a spring constant comparable to the stiffness of the mount to be tested. System verification shall be performed using springs of known spring constants at the excitation amplitudes and frequencies at which the mount is to be tested. System verification shall be performed at loads which produce spring constant measurements that bracket the expected dynamic stiffness range of the mount to be tested. This shall be accomplished by adding or removing springs within the stiffness range of intended mount measurements. Care shall be taken to assure system verification measurements are conducted within the linear load range of the springs. All spring dynamic stiffness measurements (corrected for test machine stiffness) should be within 3 percent of the known spring constants. If this is not the case, the error shall be investigated and corrected. This procedure is not intended to replace the periodic calibration of transducers, signal conditioners, and other measuring instruments as required by ASTM D5992.

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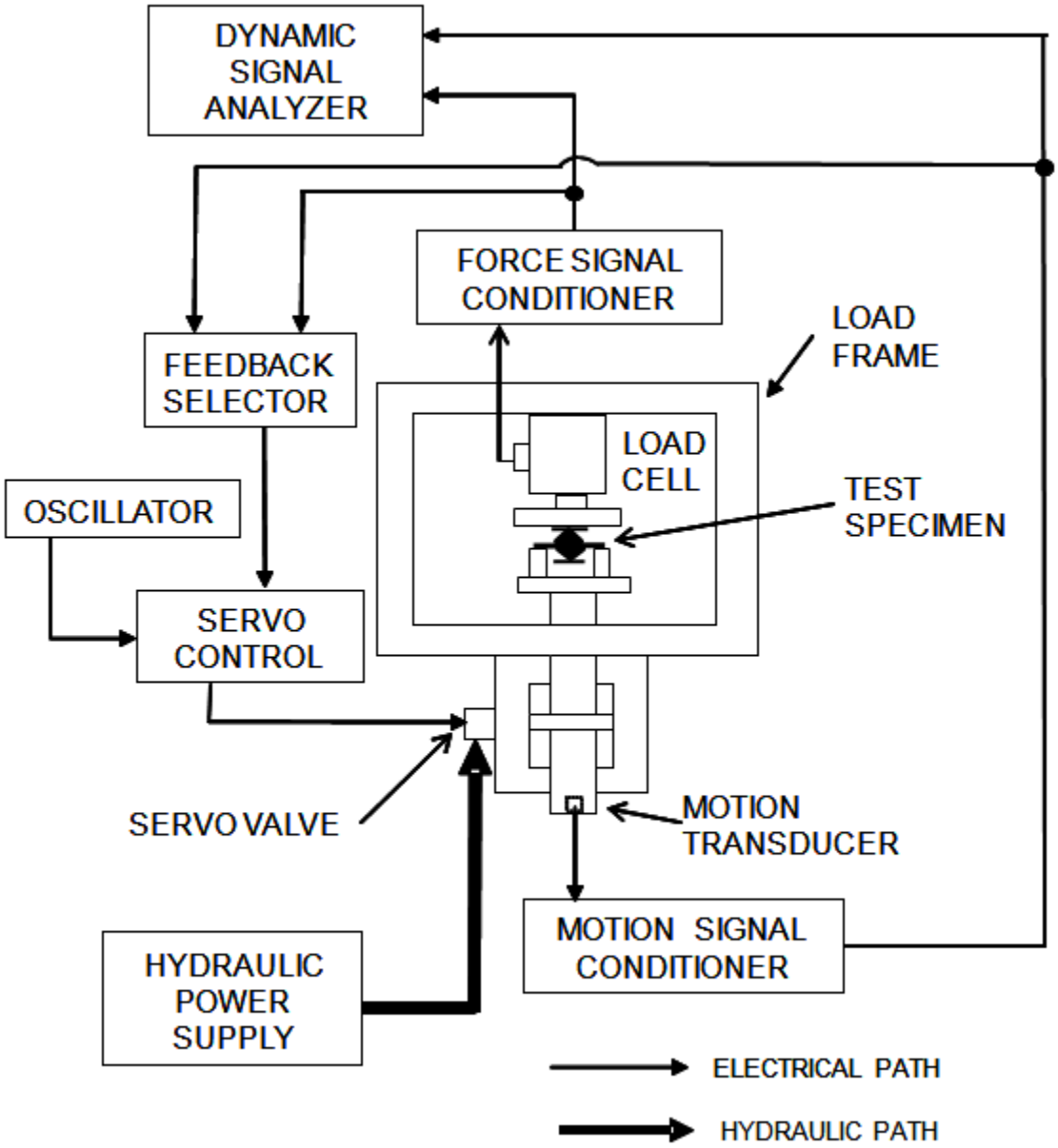


FIGURE 1. Diagram of major components of servo-hydraulic universal test machine.

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4.4.2.1.2 Method for measuring dynamic stiffness. Dynamic stiffness shall be measured for each mount in the axial direction at the test conditions specified in table XIII to determine conformance with 3.5.2. The measurement method is based on the transmitted force principle and is used to define the properties of a dynamic model consisting of a parallel combination of an ideal spring and damper to represent the mount. Although a more exact model includes a mass term, it may be omitted with negligible error at low excitation frequencies. Dynamic stiffness measurements, as defined in this specification, shall be acquired by following the guidelines given in ASTM D5992 for forced non-resonant measurements using the Fast Fourier Transform (FFT) method. This requires measurement of the sinusoidal displacement amplitude (X) at the input of the mount and transmitted dynamic force amplitude (F) at the blocked output of the mount, as well as the phase relationship (θ) between them. The dynamic stiffness is calculated from the measurements as follows:

$K = C(F/X)(\cos(\theta + \delta))$

Where "C" and "δ" are minor adjustment factors, if necessary, to account for test apparatus flexibility and instrumentation phase shift (see 4.4.2.1.1).

The measured stiffness may require adjustment to account for test apparatus flexibility and instrumentation phase shift (see 4.4.2.1.1). The mount shall be "conditioned" by cycling it three times from "0" to its upper rated load (15 pounds for 11M15, 25 pounds for 11M25, and 50 pounds for 10M50) prior to the start of measuring dynamic stiffness. Then, at each static load condition, the mount shall be vibrated and allowed to settle before readjusting (if necessary) to the required load before acquiring dynamic stiffness data.

TABLE XIII. Dynamic stiffness test conditions.

Mount type	Axial static load (lbs)	Excitation	
		(inches, pk-pk)	(Hz)
11M15	9, 12, 15	0.01	11
11M25	15, 18, 22, 25	0.01	11
10M50	25, 33, 42, 50	0.01	10

4.4.2.2 Static strength test.

4.4.2.2.1 Apparatus for conducting strength test. A suitable test system capable of loading the mounts at a constant rate of deflection while measuring load and deflection shall be used. A commonly used type of equipment for measuring load-deflection is a Universal Testing Machine. The sampling rate of measured load and deflection shall provide a sufficient number of samples to furnish smooth load-deflection plots without discontinuities. Deflection shall be measured to the nearest 0.001 inch. Flexibility of the test system (load frame, force gauge, fixtures, etc.) shall be accounted for via correction of measured mount data. A test system which is at least 100 times stiffer than the static stiffness of the mounts being tested shall not require a flexibility correction. To avoid over-correction, the stiffness of the test system shall be at least 30 times greater than the stiffness of the mounts being tested. All sensors used for measurement (force and displacement transducer) shall be calibrated at regular intervals.

4.4.2.2.2 Axial direction strength test. A single mount shall be secured in a test fixture and subjected to four loading and unloading cycles at a constant deflection rate not to exceed 0.3 inch per minute. The peak load obtained shall be to "Level A" for the first three loading cycles and to "Level B" for the fourth loading cycle (refer to table XIV). Load-deflection data shall be recorded during the fourth loading cycle. The mount shall be inspected during and after testing to determine compliance with 3.5.3

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TABLE XIV. Strength test peak loads.

Mount type	Level A (lbs)	Level B (lbs)
11M15	1000	1300
11M25	1390	1810
10M50	2170	2835

4.4.2.2.3 Radial direction strength test. Two mounts shall be tested in the radial direction as a pair while axially compressed back-to-back in a fixture similar to that shown in figure 2. Referring to the figure, axial compression is adjusted via the four threaded rods. The axial deflection of each mount shall correspond to its upper load rating as measured during the axial strength test. Mounts shall be subjected to four radial loading and unloading cycles at a constant deflection rate not to exceed 0.3 inch per minute while compressed statically in the axial direction. The peak load applied to the test fixture shall be to “Level A” for the first three loading cycles and to “Level B” for the fourth loading cycle. Load-deflection data shall be recorded during the fourth loading cycle. A per-mount average load-deflection plot shall be determined by dividing the load at each displacement by 2. The mount shall be inspected during and after testing to determine compliance with 3.5.3.

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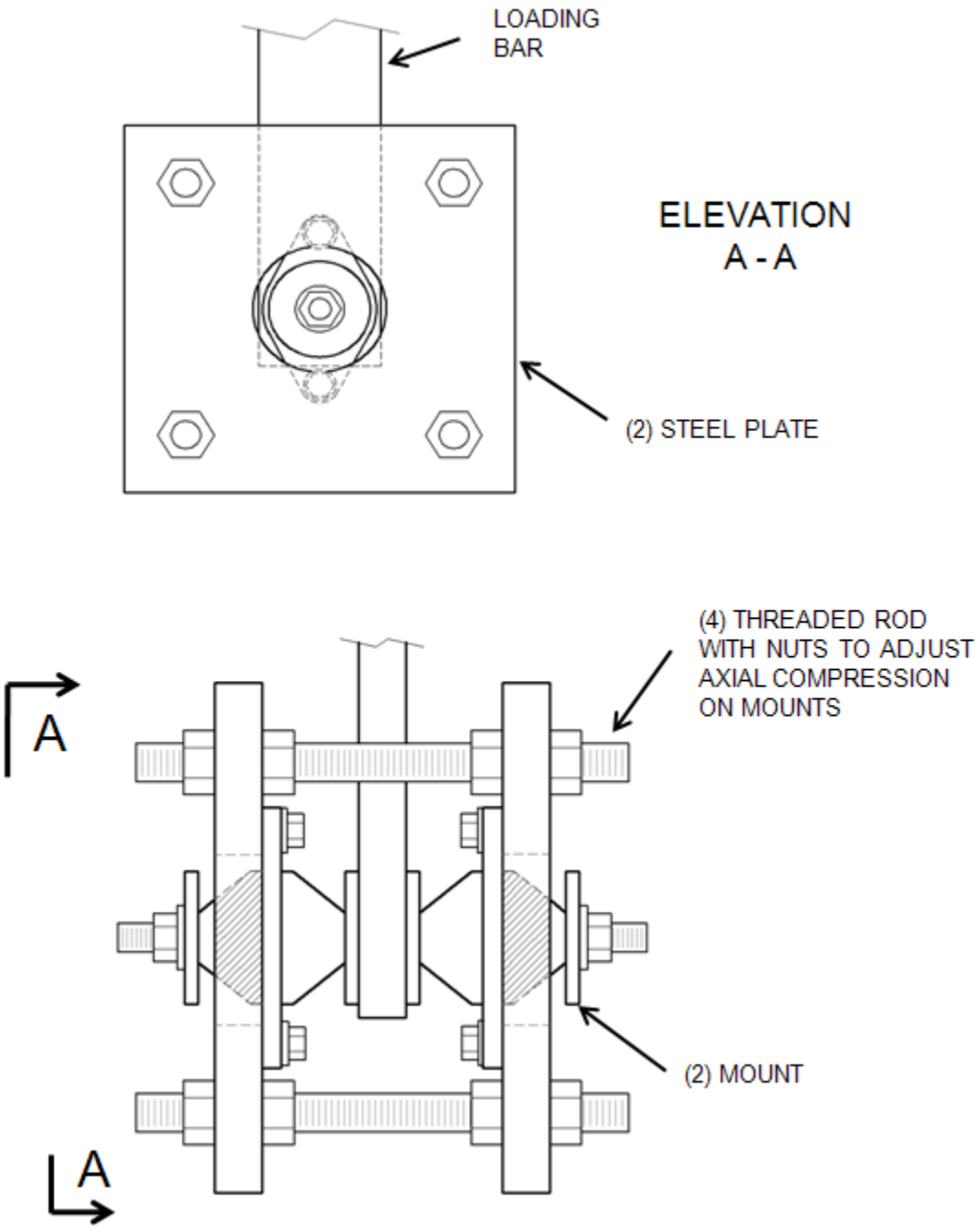


FIGURE 2. Fixture to conduct strength test in the radial direction.

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4.4.2.3 Fatigue-drift test. The test apparatus shall be capable of providing cyclical vibration to the test fixture. Refer to figure 3. All mount height measurements shall be to the nearest 0.001 inch. The mount shall be secured in a test fixture and axially loaded in compression with a mass to its upper-rated load. One hour after application of the mass, mount height shall be measured. The fixture shall then be vibrated at the natural frequency of the mount-mass system in the axial direction for at least 500,000 cycles. The amplitude of the excitation force shall ensure the mount is deflected 0.05 inch each side of the equilibrium position (0.10 inch double amplitude). The vibration tests shall be conducted at regular and equal time intervals for 4 consecutive days. The mount shall remain loaded for the entire fatigue-drift test, including the time periods between vibration intervals. Mount height shall be measured at the end and beginning of each vibration test run. A final mount height measurement shall be made 96 hours after initial static loading. Mount drift to determine compliance with 3.5.4 shall be the difference in the height measurements taken 1 hour after loading and 96 hours after loading. The test shall be conducted at an ambient temperature of 80±5 °F. NOTE: It is usually convenient to measure the amount of drift from an arbitrary datum from which the change in mount height can be determined.

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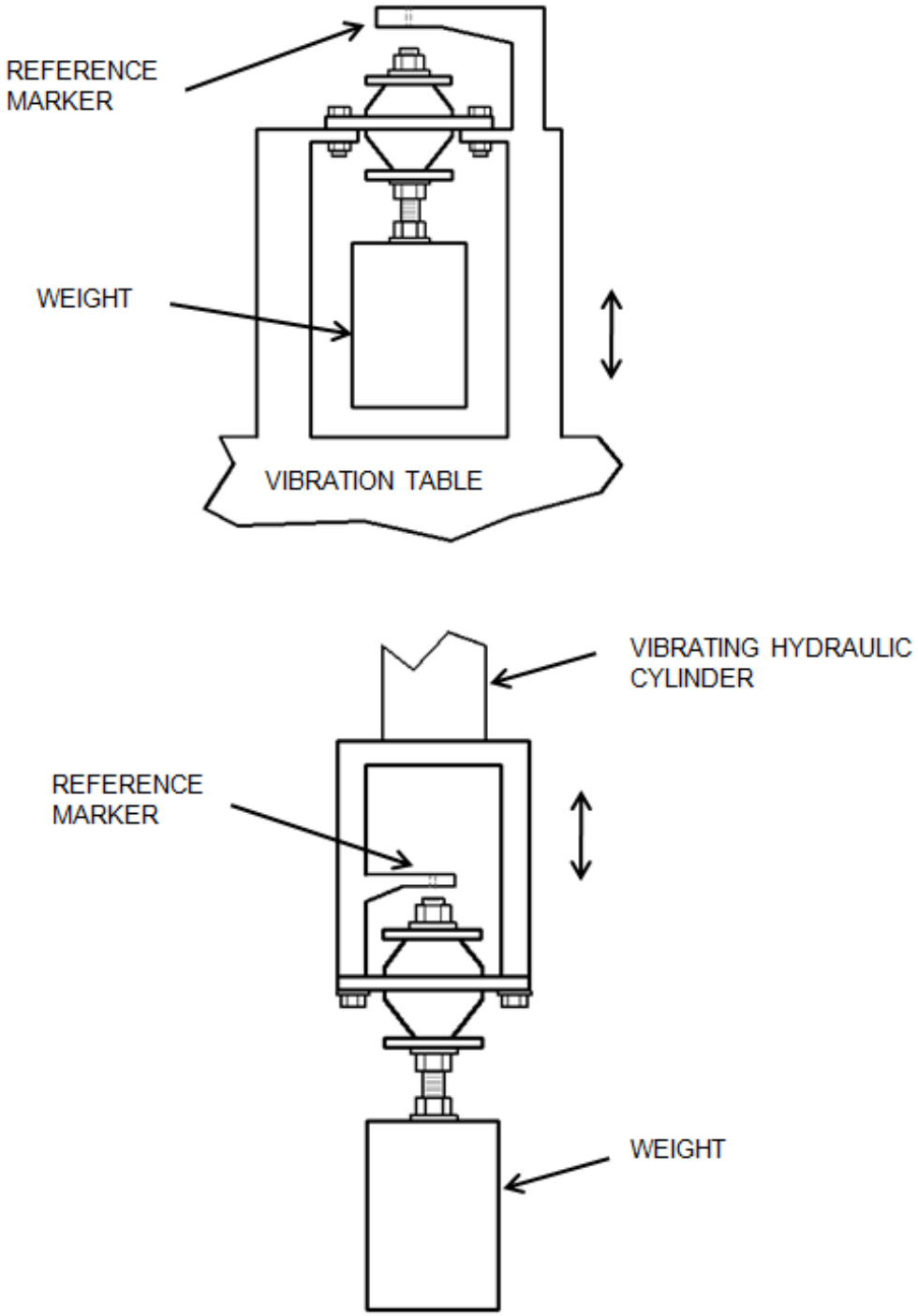


FIGURE 3. Two examples of an apparatus capable of conducting the fatigue-drift test.

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4.4.2.4 Delamination and porosity test method. The mount resilient element shall be cut into two parts and examined for porosity. The cut parts shall be immersed in Toluene (ACS reagent grade, ≥99.5 percent) for 16±¼ hours at a temperature of 80±5 °F. After removal from Toluene, the sections of the mount shall be examined to determine evidence of separation into distinct layers or laminations.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging and storage requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The mount types covered in this specification are intended to isolate shock and vibration on surface ships and submarines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Mount type (see 1.2.1).

c. Type of metal steel or manganese bronze (see 3.4.1).

d. Packaging requirements, shipping, and marking (see 5.1 and 6.6).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 19379 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Definitions.

6.4.1 Axial direction. The direction which is parallel to the center bolt axis of the mount as shown in 803-1385824.

6.4.2 Dynamic stiffness. The real part of the frequency-dependent complex ratio of force on the blocked output side of the mount to displacement on the input side during sinusoidal vibration. Also referred to as spring constant or elastic dynamic stiffness.

6.4.3 Radial direction. A direction which is perpendicular to the center bolt axis of the mount as shown in 803-1385824.

6.5 Static load-deflection, radial dynamic stiffness and damping. Approximate static deflection at rated load for each mount type is listed in table XV. Dynamic stiffness in either the radial or axial direction should be similar when subjected to a single equivalent load in that direction. Damping loss factor should be approximately 0.12.

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TABLE XV. Static deflection guidance.

Mount type	Deflection when loaded axially	Deflection when loaded radially
11M15	0.09 inch at 15 lbs	0.10 inch at 15 lbs
11M25	0.10 inch at 25 lbs	0.11 inch at 25 lbs
10M50	0.13 inch at 50 lbs	0.13 inch at 50 lbs

6.6 Marking. Waterproof marking should be provided on interior packages and exterior shipping container with the following information (see 6.2):

MOUNT TYPE:

MILITARY SPEC NO: MIL-DTL-19379

DATE OF MFG:

LOT NO:

SHELF LIFE: 7 YEARS FROM DATE OF MFG

6.7 Subject term (key word) listing.

Isolator

Shock

Vibration

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy – SH
(Project 5342-2013-002)

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